c. manganese **d. boron**

23. Sulphur is not a constituent of (AMU 2011)
a. cysteine b. methionine
c. ferredoxin **d. pyridoxine**

24. Deficiency symptoms of nitrogen and potassium are visible first in _____ (AIPMT 2014)
a. senescent leaves b. young leaves
c. roots d. buds

25. The first stable product of fixation of atmospheric nitrogen in leguminous plants is _____ (AIPMT 2013)
a. NO^{-3} b. glutamate c. NO^{-2} **d. ammonia**

26. C4 plants are more efficient in photosynthesis than C3 plants due to (AIPMT 2010)
a. presence of thin cuticle
b. lower rate of photorespiration
c. higher leaf area
d. presence of larger number of chloroplast in the leaf cells.

27. Chlorophyll b is (JIPMER 1980)
a. $C_{54}H_{70}O_6N_4Mg$ **b. $C_{55}H_{70}O_6N_4Mg$**
c. $C_{55}H_{72}O_5N_4Mg$ d. $C_{45}H_{72}O_5N_4Mg$

28. Synthesis of ADP + Pi → ATP in grana is (AIIMS 1993)
a. phosphorylation **b. photophosphorylation**
c. oxidative phosphorylation d. photolysis

29. In chloroplast, chlorophyll is present in the (AIPMT 2004)
a. stroma b. outer membrane
c. inner membrane **d. thylakoids**

30. Electrons from the excited chlorophyll molecule of photosystem II are accepted first by (AIPMT 2008)
a. quinone b. ferredoxin
c. cytochrome-b d. cytochrome-f

31. Read the following four statements A,B,C and D. Select the right option (AIPMT 2010)
A. Z scheme of light reaction takes place in the presence of PS I only
B. only PS I is functional in cyclic photophosphorylation
C. cyclic photophosphorylation results into synthesis of ATP and NADPH2
D. stroma lamellae lack PS II as well as NADP
a. A and B b. B and C c. C and D **d. B and D**

32. Photolysis of each water molecule in light reaction will yield ___ (Kerala CEE 2007)
a. 2 electrons and 4 protons
b. 4 electrons and 4 protons
c. 4 electrons and 3 protons
d. 2 electrons and 2 protons

33. Photosynthetic active radiation (PAR) has the following range of wavelength (AIPMT 2005)
a. 400-700 nm b. 450-920 nm
c. 340-450 nm d. 500-600 nm

34. Phosphoenol pyruvate (PEP) is the primary CO_2 acceptor in __ (NEET 2017)
a. C_3 plants **b. C_4 plants**
c. C_2 plants d. C_3 and C_4 plants

35. With reference to factors affecting the rate of photosynthesis, which of the following statements is not correct? (NEET 2017)
a light saturation for CO_2 fixation occurs at 10 % of full sunlight
b. increasing atmospheric CO_2 concentration up to 0.05% can enhance CO_2 fixation rate
c. C_3 plants respond to higher temperature with enhanced photosynthesis while C_4 plants have much lower temperature optimum.
d. tomato is a greenhouse crop which can be grown in CO_2 enriched atmosphere for higher yield

36. A plant in your garden avoids photorespiratory losses, has improved water use efficiency, shows high rates of photosynthesis at high temperatures and has improved efficiency of nitrogen utilization. In which of the following physiological groups would you assign this plant? (NEET PHASE I 2016)
a. C_4 b. CAM c. Nitrogen fixer d. C_3

37. Emerson's enhancement effect and Red drop have been instrumental in the discovery of (NEET PHASE I 2016)
a. two photosystems operating simultaneously
b. photophosphorylation and cyclic electron transport
c. oxidative phosphorylation
d. photophosphorylation and non-cyclic electron transport

38. The process which makes major difference between C_3 and C_4 plants is (NEET PHASE II 2016)
a. glycolysis b. calvin cycle
c. photorespiration d. respiration

39. In a chloroplast the highest number of protons are found in (NEET PHASE I 2016)
a. lumen of thylakoids
b. inter membrane space
c. antennae complex
d. stroma

40. Oxidative phosphorylation is (NEET 2016)
a. formation of ATP by transfer of phosphate group from a substrate to ADP
b. oxidation of phosphate group in ATP
c. Aaddition of phosphate group to ATP
d. formation of ATP by energy released from electrons during substrate oxidation.

41. Which of the biomolecules is common to respiration-mediated breakdown of fats, carbohydrates and proteins? (NEET 2013, 2016)
a. glucose-6-phosphate
b. fructose1,6-bisphosphate
c. pyruvic acid
d. acetyl CoA

42 Which statement is wrong for Krebs cycle? (NEET 2017)
a. there is one point in the cycle where FAD is reduced to $FADH_2$
b. during conversion of succinyl CoA to succinic acid, a molecule of GTP is synthesised.
c. the cycle starts with condensation of acetyl

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